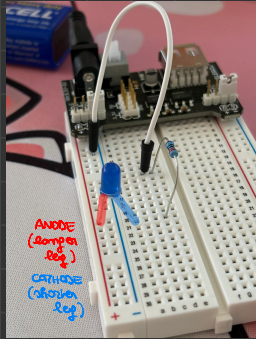
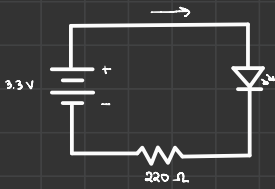
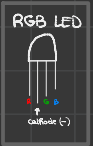


SIMPLE LED CIRCUIT



LED = Light Emitting Diode

current only flows in one direction $\rightarrow$



PROPER RESISTOR VALUE

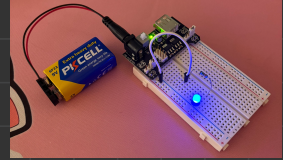
You need to know:

- What component for $\rightarrow$ LED
- Voltage of power source $\rightarrow$ 3V BATTERY
- Forward voltage of component $\rightarrow$ ~ 3V BLUE LED
- How many Amperes does the component need

$$R = \frac{V_{SUP} - V_{LED}}{I_{LED}} \quad \Omega \quad \left[R = \frac{V}{I} \right]$$

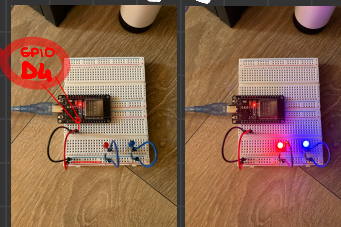
$$I = \frac{V}{R} = \frac{3.3 - 3}{220} = 1.36 \text{ mA}$$

very low $\Rightarrow$ LED not very bright



ADDING THE ESP32

```
1 const int ledPin = 4;
2
3 void setup() {
4   pinMode(ledPin, OUTPUT);
5 }
6
7 void loop() {
8   digitalWrite(ledPin, HIGH);
9   delay(1000);
10  digitalWrite(ledPin, LOW);
11  delay(1000);
12 }
```



WITH RGB LED

Gradually changing color

```
1 #define RED 2
2 #define GREEN 4
3 #define BLUE 5
4
5 void setup()
6 {
7   pinMode(RED, OUTPUT);
8   pinMode(GREEN, OUTPUT);
9   pinMode(BLUE, OUTPUT);
10 }
11
12 void setColor(int red, int green, int blue) {
13   analogWrite(RED, red);
14   analogWrite(GREEN, green);
15   analogWrite(BLUE, blue);
16 }
17
18 void loop() {
19   for (int i = 0; i < 255; i++) { setColor(255 - i, i, 0); delay(10); } // red -> green
20   for (int i = 0; i < 255; i++) { setColor(0, 255 - i, i); delay(10); } // green -> blue
21   for (int i = 0; i < 255; i++) { setColor(i, 0, 255 - i); delay(10); } // blue -> red
22 }
```

